



Enhancing Thermal Energy Storage Performance via Orientation Optimization of Cylindrical Phase Change Material Systems

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Abstract: An experimental investigation was conducted to evaluate the influence of geometric orientation on the thermal performance of a cylindrical phase change material energy storage system incorporating concentric heat transfer fluid tubes. Three configurations—vertical, inclined, and horizontal—were systematically examined to determine their effects on heat transfer characteristics, melting dynamics, and energy storage capacity. The phase change material was subjected to charging processes using heat transfer fluid inlet temperatures of 60°C, 70°C, and 80°C at a constant mass flow rate of 1 kg/min, while discharging experiments were performed under identical flow conditions to ensure consistency. It was observed that the melting time was significantly reduced in the vertical configuration, exhibiting a decrease of 29.2% compared to the horizontal arrangement and 19.4% relative to the inclined configuration at an inlet temperature of 80°C. This enhancement was attributed to the intensification of natural convection within the molten phase change material region. Furthermore, at a charging duration of 140 minutes, the total thermal energy stored in the vertical configuration was found to be approximately 4.4%, 17.4%, and 19.4% higher than that of the horizontal configuration for heat transfer fluid inlet temperatures of 60°C, 70°C, and 80°C, respectively. The results demonstrate that the optimization of system orientation plays a critical role in enhancing both the charging rate and storage capacity of phase change material-based thermal energy storage systems. These findings provide valuable design insights for the development of high-efficiency latent heat storage units in renewable energy and thermal management applications.

Keywords: Phase change material; Heat transfer fluid; Thermal energy storage; Natural convection; Melting time; Discharge efficiency

1 Introduction

A thermal energy storage system is a device that collects, stores, and uses peak daytime energy. Phase change materials' latent heat capacity makes them useful for thermal energy storage. Thermal energy storage has been the subject of numerous investigations by researchers [1–3]. Additionally, many solar collectors have been researched to improve their thermal performance [4, 5]. In the field of thermal energy storage, paraffin wax is the most studied phase change substance [6, 7]. Phase change materials have also been investigated for use in high-temperature solar receivers to increase solar collector productivity [8–11]. Additives and encapsulation techniques have been used by other researchers to improve energy storage efficiency [12, 13]. Senthil et al. [14] studied the melting behavior of the phase change material (RT-50) within a horizontal annular space formed between an isothermal inner tube and an insulating outer casing. The melting rate was affected by the inlet temperature, increasing proportionally with increasing inlet temperature.

Senthil et al. [15] conducted experiments and computed the behavior of the phase-change melting process within a horizontal annular container. The study focused on the effect of inlet temperature on both the melting process and the onset of natural convection. Increasing the temperature of the phase-change liquid causes it to move upwards,

while a cold phase-change liquid moves downwards. The study involved numerical simulations of the charging and discharging processes in a rotating latent heat storage unit that contains a phase change material within the vertical and horizontal annular spaces between the casing and the tube. A lower rotational speed of the latent heat storage unit results in greater heat transfer and melting rate improvement compared to a higher rotational speed [16]. Previous studies have shown the need for more comprehensive experimental studies. Therefore, this study aims to conduct an experimental study that clearly demonstrates how container orientations (vertical/inclined/horizontal) affect the melting rate, energy storage, and temperature distribution with the use of phase change materials with enhanced conductivity (paraffin wax) and to identify the “most efficient” orientation not only in melting but also in the charging and discharging cycles.

2 Materials and Methods

The experimental setup consists of a cylindrical phase change material (heat storage container), a pump, a heater with temperature control, temperature gauges, flow valves, and a piping system. The schematic diagram of the test is shown in Figure 1. These heat transfer fluids are also made of copper and are 1000 mm long. The three orientations are examined in thermal energy storage systems with constant flow rates of 1 kg/min and constant temperatures of 60°C, 70°C, and 80°C. These factors are considered in the design and construction of the thermal energy storage system to achieve enhanced charging performance and improved energy storage efficiency. To hold the heat from hot water, paraffin wax is placed inside a copper cylindrical container. The phase change material container has a concentrically fitting copper tube. Valves are used to regulate water flow. The phase change material container is kept at a nearly constant temperature by the use of a 10-kW electric heater with thermostat temperature control. A rotameter is used to determine the flow rate. Three K-type thermocouples installed along the shell's length are measured and recorded. Table 1 shows the physical properties of the phase change material and the container dimensions.

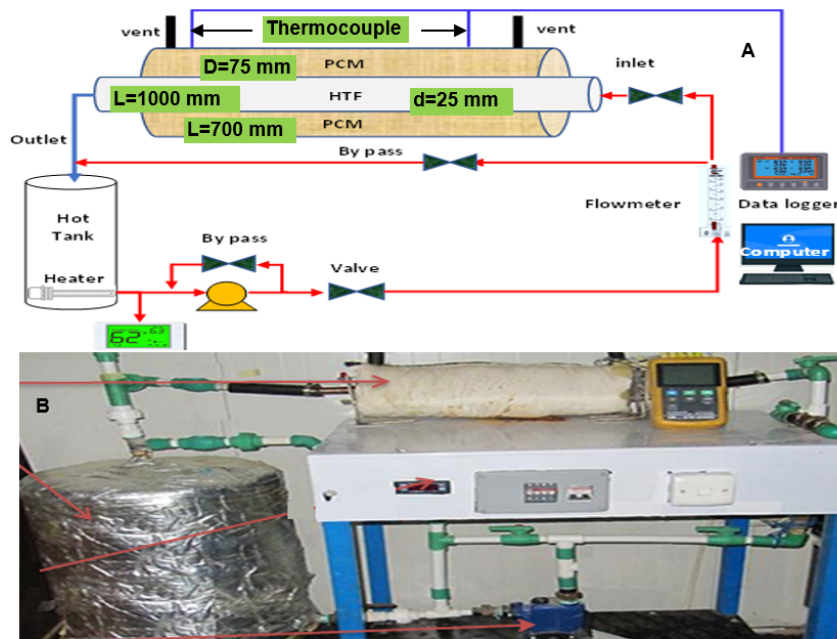


Figure 1. A: Schematic diagram of the experimental setup; B: photograph of the phase change material annular cavity heat exchanger

Note: PCM = phase change material; HTF = Heat transfer fluid.

3 Data Reduction

Water flows through a copper tube to complete the discharge process. The heat is recovered and transferred to the cold water passing via the copper tube and phase change material. Following the charging process, the phase change material storage unit is subjected to three cooling tests under ambient settings. The duration of heat storage is determined by monitoring the phase change material temperature while the insulated phase change material container is kept at room temperature and allowed to cool momentarily. Eq. (1) uses the specific heat of water and the temperature differential between the heat transfer fluid's inlet and output temperatures to determine how much heat is transmitted by the hot water.

Table 1. Dimensions and physical property [17–19]

Thermal Storage Unit	Dimensions (mm)
Diameter tube	75
Thickness	2.5
Length	1000
Heat Transfer Fluid Tube	Dimensions (mm)
Diameter tube	25
Thickness	1
Length	700
Phase Change Material (Paraffin Wax, RT-42)	Physical Property
Heat storage capacity	165 kJ/kg
Thermal conductivity	0.2 W/m·K
Density (solid)	880 kg/m ³

The amount of heat (W) produced by the water is calculated, as shown in Eq. (1) [20].

$$Q_{\text{input water}} = \dot{m}C_p (T_{\text{in}} - T_{\text{out}}) \quad (1)$$

where, $\dot{m}$ and C_p are the mass flow rate (kg/s) and the specific heat of the heat transfer fluid (kJ/kg·K). The heat generated by the phase change material during the time interval Δt is calculated, as shown in Eq. (2) [20].

$$Q_{\text{inputPCM}} = \sum \dot{m}C_p (T_{\text{in}} - T_{\text{out}}) \cdot \Delta t \quad (2)$$

Eq. (3) describes the sensible heat of the solid and liquid, as well as the latent heat, in the total stored heat energy [21].

$$Q_{\text{storage}} = m \left[\int_{T_i}^{T_m} m \cdot C_{p,\text{liquid}} \cdot dT + m \cdot \Delta h_m + \int_{T_m}^{T_f} m \cdot C_{p,\text{liquid}} \cdot dT \right] \quad (3)$$

where, T_i and T_m are the initial and melting temperatures, respectively, and m and C_p are the mass and specific heat of the storage material.

4 Uncertainty Analysis

Uncertainty analysis was used to confirm experimental values and identify potential sources of error in data measurement tools. Using measured data such as temperature, mass flow rate, and thermophysical characteristics, the convective heat transfer was calculated. The calculated uncertainties are presented in Table 2.

Table 2. Error analysis [20, 21]

Parameters	Uncertainties (%)
Temperature	±0.2
Mass flow rate	±0.5
Rate of heat transfer (Q)	±0.35

5 Results and Discussion

Figure 2 illustrates the liquid fraction (the ratio of melted phase change material to the total amount) as a function of time (in minutes) for three different heat transfer fluid temperatures (60°C, 70°C, and 80°C). Increasing the heat transfer fluid temperature roughly halves the total melting time (from 260 to 130 minutes). This demonstrates that the melting process in this specific vertical heat exchanger is highly sensitive to the temperature of the heat source. A vertical orientation in a cylinder leads to faster melting of a phase change material compared to a horizontal orientation, primarily due to the mechanism of natural convection. As the material begins to melt, the hot (less dense) liquid rises to the top, while the cold liquid sinks to the bottom. In a vertical orientation, the material has a longer path of movement along the wall of the heated cylinder.

Figure 3 illustrates the relationship between energy stored (kJ) and time (min) for three different heat source temperatures (60°C, 70°C, and 80°C). Increasing the water temperature significantly reduces the time required to

charge the phase change material system. In terms of efficiency, 80°C is the most effective operating temperature. The vertical position in the cylinder enhances heat transfer, and this continuous vertical movement of the fluid effectively “scrapes” heat from the wall and transfers it to the remaining solid part. In short, the vertical position creates stronger and more efficient convection currents covering a larger surface area of the solid.

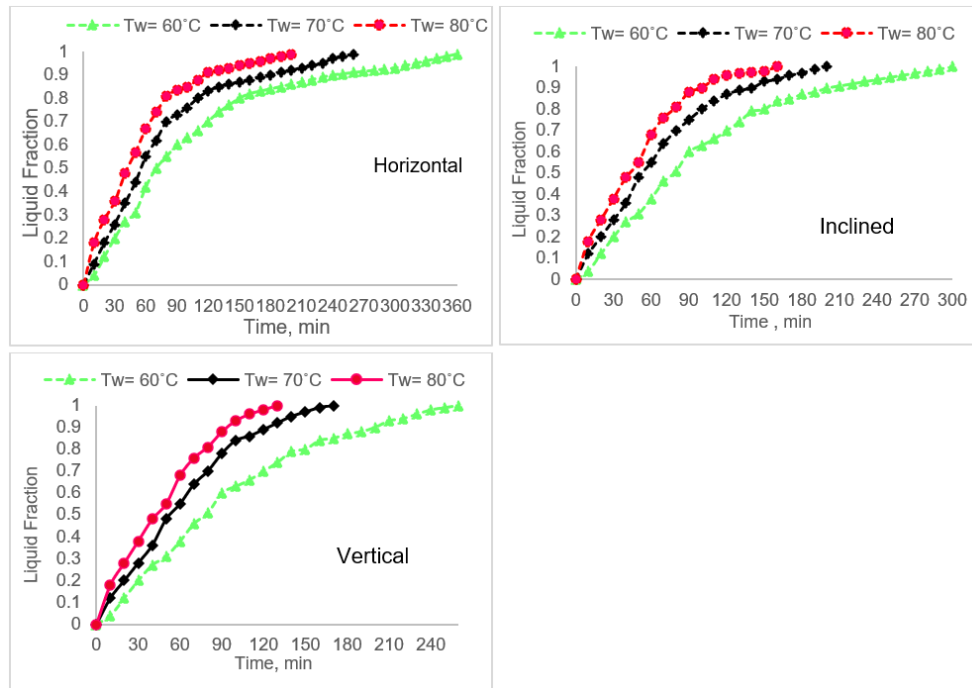


Figure 2. Variation of liquid fraction of the phase change material with time at different heat transfer water temperatures (T_w)

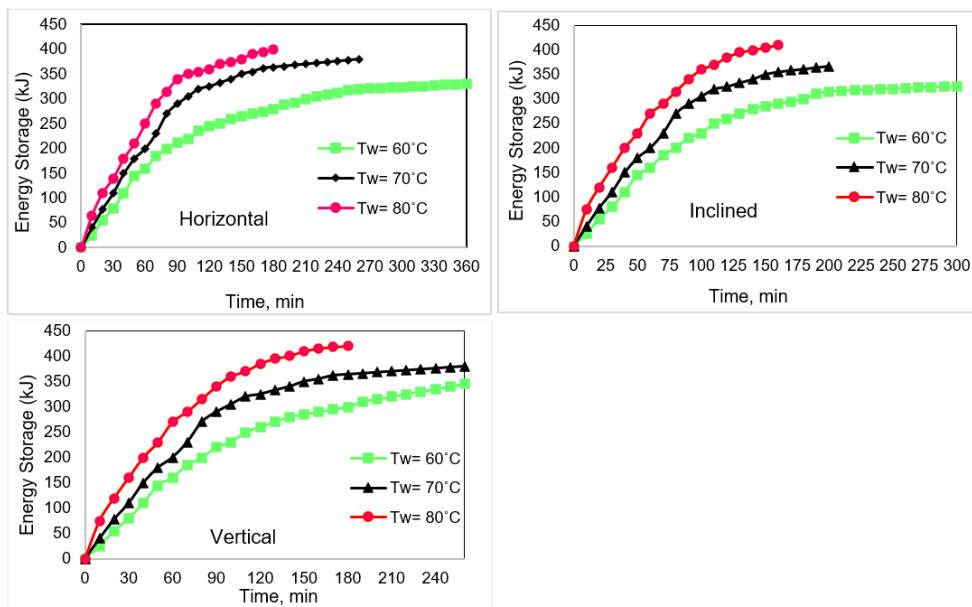


Figure 3. Variation of energy stored in the phase change material with time at different heat transfer water temperatures (T_w)

The bar chart (Figure 4) compares the heat energy stored in a phase change material unit versus the heat retrieved by water for three different container orientations: horizontal, inclined, and vertical. In the vertical position, the hot fluid accumulates at the top and comes into contact with a larger area of the solid, accelerating the melting of the entire upper portion. The melt then flows downwards. In contrast, in the horizontal position, the fluid may

concentrate at the top, leaving the lower part of the steel thermally insulated by the stagnant fluid layer.

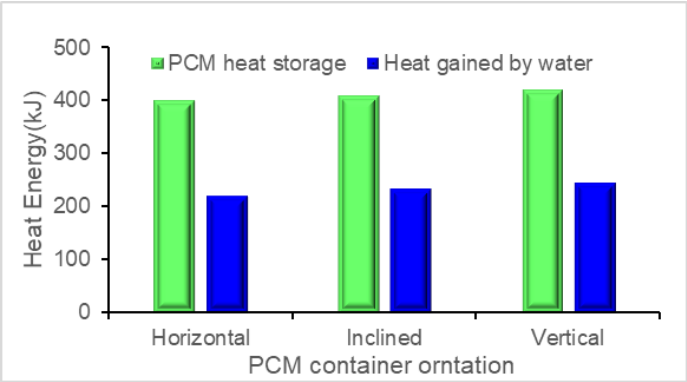


Figure 4. Energy stored in phase change material and heat gain of heat transfer fluid at room temperature at 1 kg/min
Note: PCM = phase change material.

Figure 5 displays the melting time of a phase change material as a function of the heat transfer fluid inlet temperature for three different orientations: horizontal, inclined, and vertical. To achieve the fastest charging (melting) of this specific thermal energy storage system, a vertical orientation combined with a higher heat transfer fluid inlet temperature is the most effective configuration.

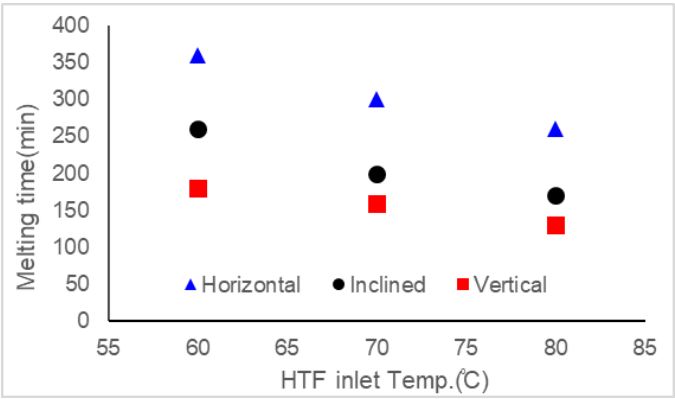


Figure 5. Melting time of a 1 kg phase change material under varying inlet temperatures (60°C, 70°C and 80°C) and container orientations
Note: HTF = heat transfer fluid.

6 Conclusion

For both heat exchanger orientations, raising the water’s inlet temperature shortened the melting time. For a horizontal heat exchanger, it decreased by roughly 27.5% and 46.3% when the inflow water temperature was raised (60°C to 70°C and 70°C to 80°C), respectively. Melting periods were shortened by 32.6% and 50.2% for vertical heat exchangers, respectively, for the same temperature rise. Compared to the horizontal heat exchanger, the vertical heat exchanger achieved a higher phase-change material melting rate. Additionally, at inlet water temperatures of 60°C, 70°C, and 80°C, the vertical heat exchanger required 28.7%, 30.4%, and 31.2% less time for the melting process than the horizontal heat exchanger. For water temperatures of 60°C, 70°C, and 80°C, respectively, with a duration of 140 minutes, the total energy stored by the phase change material during the melting process in the vertical heat exchanger was approximately 4.4%, 17.4%, and 19.4% greater than that achieved in the horizontal orientation of the heat exchanger. Therefore, vertical phase change material containers are better than horizontal ones.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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